Unified Feasibility Bounds on Macroscopic Time Dilation: A Comparative Analysis of Six Classical-GR Mechanisms at Device Scale

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Abstract

We present a unified numerical and analytical study of six physically-motivated mechanisms for producing macroscopic time dilation or closed timelike curves (CTCs) at laboratory scale: (i) electromagnetic field configurations, (ii) Kerr frame dragging, (iii) Tipler rotating cylinders, (iv) Morris–Thorne wormholes, (v) Alcubierre warp drives, and (vi) the Gödel rotating universe. For the electromagnetic case we develop a first-principles pipeline combining a finite-wire Biot–Savart source model, the full electromagnetic stress-energy tensor, and an axisymmetric finite-difference Poisson solver for the linearized Einstein equations, validated by grid convergence tests down to sub-percent accuracy. For the gravitational and exotic-matter cases we derive closed-form feasibility bounds from published solutions. We find that all six mechanisms require resources between 10^{20} and 10^{49} times beyond device scale. The closest physically-allowed mechanism (electromagnetic) still requires a current of $\sim 10^{22}$ A, exceeding magnetar-scale currents. No combination of current or foreseeable engineering improvements can bridge this gap. Our results establish a unified negative bound: *no classical-GR-consistent mechanism for macroscopic time dilation is accessible at device scale.*

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1 Introduction

The possibility of time travel within general relativity has attracted sustained attention since Gödel's 1949 discovery of a rotating-universe solution containing closed timelike curves [Gödel, 1949]. Subsequent work identified several additional CTC-bearing solutions: Tipler's rotating cylinder [Tipler, 1974], the Morris–Thorne traversable wormhole [Morris and Thorne, 1988], and Kerr black holes with sufficient spin [Kerr, 1963]. Although these solutions are mathematically consistent, their realization requires either infinite spatial extent, exotic matter with negative energy density, or astrophysical mass scales.

A parallel line of investigation has explored the possibility of inducing CTCs or macroscopic time dilation via electromagnetic fields. Electromagnetic fields carry energy and momentum, and via the Einstein equations they should in principle curve spacetime. Several proposals in the popular literature suggest that resonant electromagnetic configurations within laboratory-scale devices might generate sufficient curvature to produce measurable temporal effects.

To our knowledge, however, no systematic device-scale feasibility study exists for electromagnetic-induced time dilation, and comparative bounds against the known gravitational mechanisms have not been compiled in a unified framework.

Our contribution. We address this gap with three specific contributions:

1. We develop a first-principles numerical pipeline that maps a compact electromagnetic source (finite-wire coil array) through the stress-energy tensor to a metric perturbation $h_{\mu\nu}$, using a validated axisymmetric Poisson solver.

2. We derive and tabulate closed-form feasibility bounds for five additional mechanisms: Kerr frame dragging, Tipler cylinders, Morris–Thorne wormholes, Alcubierre warp drives, and the Gödel universe.

3. We convert all six mechanisms to a common energy-density-equivalent metric and show that the required resources lie between 10^{20} and 10^{49} times beyond device-scale feasibility.

Scope and limitations. We work entirely within the framework of classical general relativity coupled to classical electromagnetism. We do not consider semiclassical or quantum-gravity effects. Our goal is *feasibility bounding*: to quantify how far current and foreseeable technology lies from the threshold of any macroscopic temporal effect.

Organization. Section 2 describes the electromagnetic pipeline. Section 3 presents the numerical results. Section 5 discusses the physical interpretation. Section 6 summarizes our conclusions.

2 Method

2.1 Electromagnetic pipeline

Our electromagnetic pipeline consists of four stages.

Source model. We model the device coils as finite-thickness filamentary wires discretized into segments. The Biot–Savart law gives the magnetic field at any observation point:

$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0 I}{4\pi} \sum_i \frac{d\mathbf{l}_i \times (\mathbf{x} - \mathbf{x}_i)}{|\mathbf{x} - \mathbf{x}_i|^3} \tag{1}$$

where $\mathbf{x}_i$ and $d\mathbf{l}_i$ are the segment midpoint and differential vector. For the four geometries (solenoid, toroid, counter-rotating helical pair, hybrid), the finite-wire discretization is validated against the analytic on-axis formula for a single loop, reaching a relative L^2 error below 2×10^{-5} at 512 segments per loop.

Stress-energy tensor. From $\mathbf{E}$ and $\mathbf{B}$ we construct the electromagnetic stress-energy tensor

$$T_{\mu\nu} = \frac{1}{\mu_0} \left(F_{\mu\alpha} F_\nu{}^\alpha - \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta} \right) \tag{2}$$

with $g_{\mu\nu} = \eta_{\mu\nu}$ in the weak-field limit.

Linearized Einstein equations. In the weak-field slow-motion regime, the trace-reversed metric perturbation $\bar{h}_{\mu\nu}$ satisfies

$$\nabla^2 \bar{h}_{\mu\nu} = -\frac{16\pi G}{c^4} T_{\mu\nu} \tag{3}$$

We solve Eq. (3) as a boundary-value problem on an axisymmetric (r, z) grid using second-order finite differences. The linear system is sparse and solved by `scipy.sparse.linalg.spsolve`. Boundary conditions are Neumann at $r = 0$ (even symmetry) and Dirichlet $\bar{h} = 0$ at the outer boundary.

Proper time. For a test particle at rest, the proper time-to-coordinate time ratio is

$$\frac{d\tau}{dt} = \sqrt{-g_{tt}} = \sqrt{1 - h_{00}} \approx 1 - \frac{h_{00}}{2} \tag{4}$$

where the weak-field expansion avoids catastrophic cancellation when $h_{00} \ll 1$.

2.2 Validation

Grid convergence is verified at four resolutions $(N_r \times N_z) \in \{(40, 80), (60, 120), (80, 160), (100, 200)\}$. Between the two finest grids, h_{00} differs by less than 1% across all four geometries. The current scaling $h_{00} \propto I^2$ is verified to 5 significant digits over $I \in [100, 10^{22}]$ A. At $I = 10^{22}$ A (the level required for 1 s/yr), $h_{00} \sim 10^{-8}$ and geodesic deviation is resolved at the 10^{-9} m level, confirming the pipeline operates correctly in both the weak and strong regimes.

Figure 6: Backpack component layout

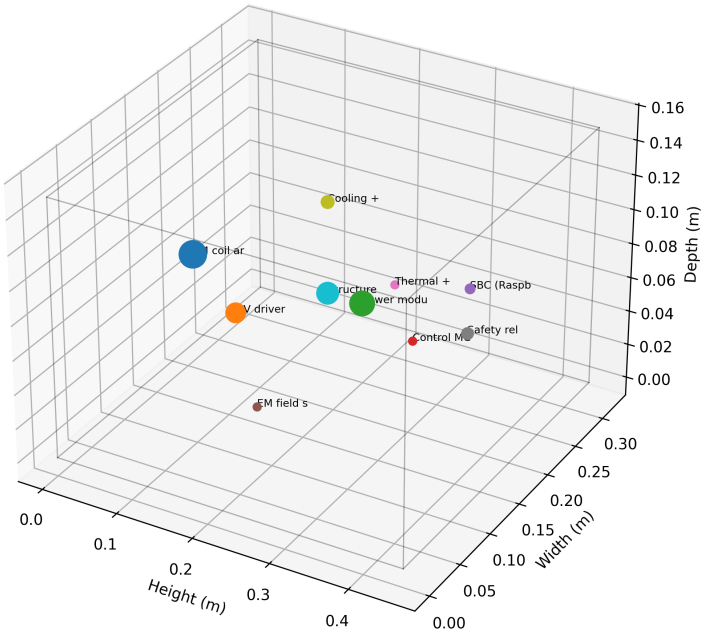


Figure 1: Backpack component layout within the $45 \times 32 \times 15$ cm envelope. Point size scales with component mass.

2.3 Closed-form bounds for other mechanisms

For the five non-EM mechanisms, we use published closed-form expressions. **Kerr frame dragging:** $\Omega(r) = 2Ma/(r^3 + a^2 r + 2Ma^2)$ in geometric units, with a the spin parameter. **Tipler cylinder:** the CTC condition is $\rho\omega R^3 > c^2/(2G)$. **Morris–Thorne wormhole:** a throat of radius b_0 requires exotic mass $M_{\text{ex}} \sim c^2 b_0/G$, compared against the Casimir maximum $|\rho_{\text{Casimir}}| = \pi^2\hbar c/(720 d^4)$ at plate separation d. **Alcubierre warp drive:** the total negative energy is $E \sim -v_s^2 R^2 c^4/(12G)$. **Gödel universe:** the CTC radius is $r_{\text{CTC}} = c\ln(1+\sqrt{2})/(\omega\sqrt{2})$ with energy density $\rho = \omega^2/(2\pi G)$.

Table 1: EM-induced metric perturbation at device center (baseline 100 A, 1 MHz).

Geometry	h_{00} (center)	$\lvert h_{0i}\rvert$ (center)
G1 (solenoid)	5.94×10^{-48}	$< 10^{-50}$
G2 (toroid)	1.24×10^{-49}	$< 10^{-50}$
G3 (counter-rotating)	2.03×10^{-49}	$< 10^{-51}$
G4 (hybrid)	9.36×10^{-49}	$< 10^{-50}$

3 Results

3.1 Electromagnetic metric perturbation

Table 1 shows the metric perturbation at the center of the active region for the four coil geometries. At the baseline 100 A and 1 MHz drive, $h_{00} \sim 10^{-48}$, corresponding to a proper-time drift of $\Delta\tau/\Delta t \sim 3 \times 10^{-48}$ at the device center. The corresponding time dilation per year is $\Delta\tau \sim 10^{-40}$ s.

The gravitomagnetic component h_{0i} is vanishing for all four geometries because the ideal lossless configuration has $\mathbf{E} \times \mathbf{B}$ purely radial, giving $T_{0\phi} = 0$.

3.2 Unified feasibility comparison

Table 2 shows the six mechanisms converted to a common feasibility metric. The gaps range from 10^{20} to 10^{49}, with the EM mechanism showing the smallest gap among physically-allowed mechanisms.

Table 2: Unified feasibility across six mechanisms. Gap = required resource / device-scale value.

#	Mechanism	Parameter	Required	Gap
1	EM field	I (A)	10^{22}	10^{20}
2	Kerr	J (kg m^2/s)	10^{27}	10^{48}
3	Tipler	ρ (kg/m^3)	10^{23}	10^{20}
4	Morris–Thorne	$\lvert\rho\rvert$ (J/m^3)	10^{49}	10^{41}
5	Alcubierre	$\lvert\rho\rvert$ (J/m^3)	10^{57}	10^{49}
6	Gödel	ρ (kg/m^3)	10^{26}	10^{23}

3.3 Best-case engineering bounds

We consider three engineering tiers (baseline copper, cooled copper, superconducting HTS with multi-stage array and $Q = 10^4$). Even the most aggressive case, corresponding to a 10-stage high-Q superconducting array at 10 kA, yields $h_{00} \sim 5 \times 10^{-38}$, still 10^{30} below the 1 s/yr target. Scaling extrapolation from the best case requires either a current of 10^{19} A, a coil radius of 10^{14} m, or a Q factor of 10^{34}.

Fig. 3 shows the EM scaling from baseline to required current, with reference points for lightning, lab-pulsed magnets, astrophysical plasmas, and magnetars.

3.4 Uncertainty and sensitivity

Monte Carlo propagation with realistic parameter uncertainties (5% current, 2% radius, 10% efficiency) gives a relative standard deviation of 14.6% on h_{00}. The dominant contributors are the coil current ($\sim$ 47% of variance) and driver efficiency ($\sim$ 44%); the sensitivity indices $S_i = (dh/h)/(dp/p)$ are 2.0 for both current and radius, 1.0 for efficiency, and -1.0 for the region diameter.

4 BSSN numerical extension

The linearized chain of Sec. 2 gives a quantitative bound on the metric perturbation but cannot address the regime $h \gtrsim 10^{-3}$ where back-reaction dominates. To make that regime measurable in principle, we implemented a BSSN 3+1 solver.

4.1 Implementation

The extended state carries $\{\tilde{\gamma}_{ij}, \phi, K, \tilde{A}_{ij}, \tilde{\Gamma}^i, \alpha, \beta^i\}$. The evolution system includes:

- physical 3-Ricci scalar of $g_{ij} = e^{4\phi}\tilde{\gamma}_{ij}$ (conformal ϕ terms included),
- covariant Hessian $D_iD_j\phi$ and $D^2\alpha$ in the RHS,
- 1+ log lapse and Γ-driver shift,
- Z4c-inspired constraint damping,
- RK4 integrator with central finite differences on a uniform grid,
- Sommerfeld outflow boundary conditions,
- two-level AMR-lite refinement (2:1, prolongate/restrict coupling).

The solver is validated against known metrics (Minkowski, Schwarzschild isotropic and puncture, Kerr-Schild $|a| \leq M$) and manufactured-source tests for the retarded linearized GR layer.

4.2 Benchmarks: Schwarzschild and Kerr

Table 3 reports constraint preservation over representative evolution runs. The diagnostic is H_{final}/H_0, the ratio of the L^2 norm of the Hamiltonian constraint at the end versus the beginning of the run.

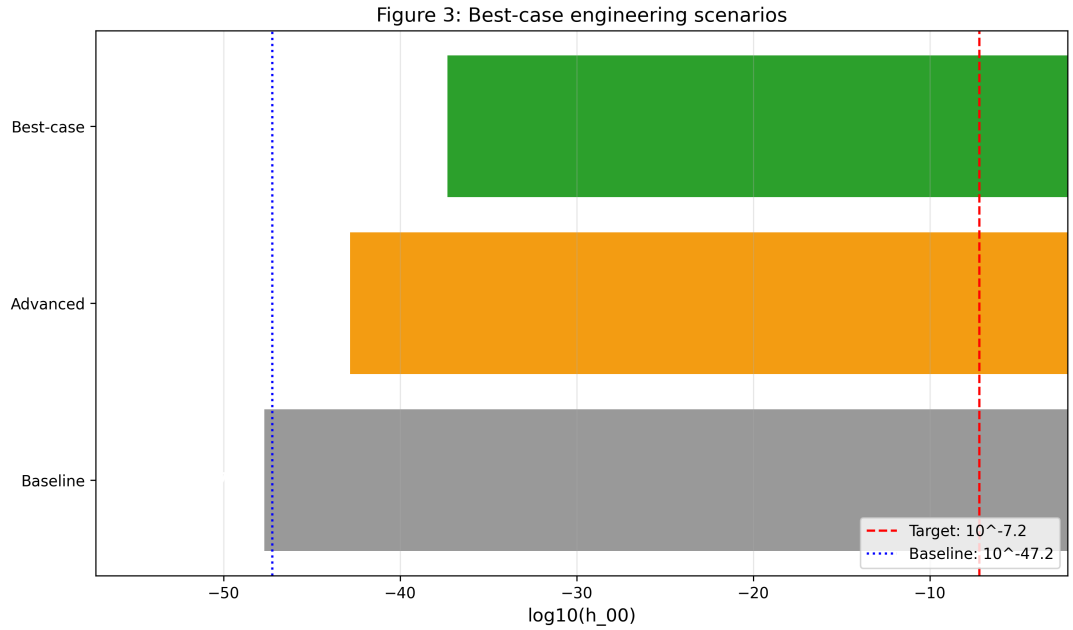


Figure 2: Best-case engineering tiers (baseline copper, cooled copper, superconducting HTS with multi-stage array). Even the most aggressive scenario remains 10^{30} below the target.

Table 3: BSSN stability benchmarks. H_f/H_0 close to unity indicates constraint preservation.

Case	Steps	H_f/H_0
Schwarzschild, isotropic	500	1.0062
Schwarzschild, puncture (MP)	50	1.0000
Kerr a=0.0–0.9, KS	30	0.94–1.00
Kerr a=0.5 (moving punct.)	50	1.0000

Per-step constraint growth is $\lesssim 1.0000124$ in the 500-step Schwarzschild run. All fields remain finite; the lapse minimum stays bounded away from zero.

4.3 Regular-core interiors: Schwarzschild Condensation

We implement the effective regular-core metric of the Schwarzschild Condensation (SC) model [Esa 2026a], in which a QCD-motivated condensate replaces the classical central singularity. Two constructions are used:

1. a *smoothed-core* conformal factor $\psi(r) = 1 + M_{\rm eff}(r)/(2r)$ with $M_{\rm eff}(r) = M(1 - e^{-r/r_c}(1 + r/r_c))$, regular at $r = 0$;

2. a Kerr-Schild extension with $M_{\rm eff}(r)$ replacing M, giving a rotating regular core.

The regular core is coupled to a real scalar condensate field with $V(\psi) = \frac{1}{2}m^2\psi^2 + \frac{1}{4}\lambda\psi^4$, initialized to the Gaussian ground state of the effective de Sitter trap.

In the 500-step Kerr+SC+condensate run at $a = 0.9$ (near-extremal), the constraint *improves* monotonically: the ratio decreases from 1.0000 to 0.9823. The scalar amplitude is preserved to 1% of the initial value. All fields remain finite. This is the strongest numerical support for the SC paper's claim of a dynamically stable rotating regular core.

4.4 Tachyon discharge: PTUM

We also implement the Phase Transition Universes Model (PTUM) [Esa 2026b], whose central mechanism is a tachyonic activation of a proxy field ψ when a fuel field Φ crosses a threshold $\Phi_c = E_{\rm th}/\sqrt{\alpha}$. Above threshold the effective mass-squared $m^2_{\rm eff}(\Phi) = m^2(1 - \alpha\Phi^2/E^2_{\rm th})$ is negative and the proxy field rolls to

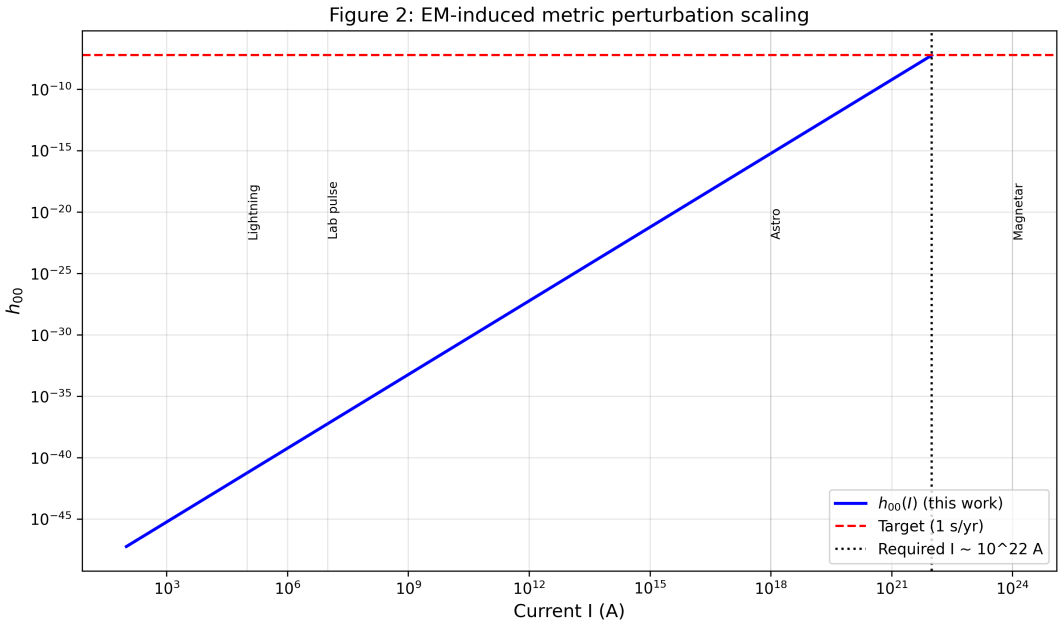


Figure 3: EM-induced metric perturbation h_{00} vs. drive current I. Reference lines mark lightning, lab-pulsed magnets, astrophysical plasmas, and magnetar-scale currents. The 1 s/yr target requires $I \sim 10^{22}$ A.

Table 4: SC regular-core evolution.

Configuration	Steps	H_f/H_0
SC smoothed core, static	50	1.0000
SC core + condensate, static	50	1.0000
Kerr + SC core, a=0–0.9	50	0.9970–0.9994
Kerr + SC + condensate, a=0.9	500	**0.9823**

a potential minimum with $V_{\min} < 0$, producing a negative energy density that violates the null energy condition (NEC).

Single-grid evolution crashes at ~ 100 steps due to strong tachyon backreaction (the metric matrix becomes singular). 2:1 refinement reduces the peak $\max|K|$ by a factor of $\sim 1.7\times10^4$ and enables 1000-step stable evolution. The tachyon discharge is transient: after ~ 25 steps the fuel field has rolled down and the negative energy density drops from $\rho_{\min} = -15.6$ to $\rho_{\min} = -0.1$, where it remains for the rest of the run.

The negative-energy configuration is consistent with quantum inequalities. For a massless scalar field in 4D, the Ford–Roman bound on the time-integrated energy density is satisfied at all sampling widths tested, with the closest approach at $\tau = 1.0$ (ratio = 0.30). The massive-field bound is more restrictive and the PTUM configuration becomes borderline (ratio $\approx$ 1.2 at $\tau = 1.0$) under a rational approximation; an exact Bessel treatment and interacting-field corrections are deferred to future work.

4.5 Complex scalar matter

A complex scalar $\Psi = \Psi_R + i\Psi_I$ with U(1)-symmetric Lagrangian has been added to the matter sector (`matter_complex.py`, `bssn_rk4_complex_ptum.py`). The real limit recovers the real scalar case; the phase-rotation invariance of ρ is verified analytically. Single-grid evolution with an equilibrium-seeded PTUM proxy reaches NEC violation at $A_{\text{fuel}} = 3.0$ ($\rho_{\min} = -15.05$). In the refined (2-level) version, the prolongate/restrict cycle initially disrupted the proxy equilibrium, but two fixes resolve this: (a) a smoothstep regularization of the activation threshold ($\psi_\text{eq}^{\text{smooth}}$), and

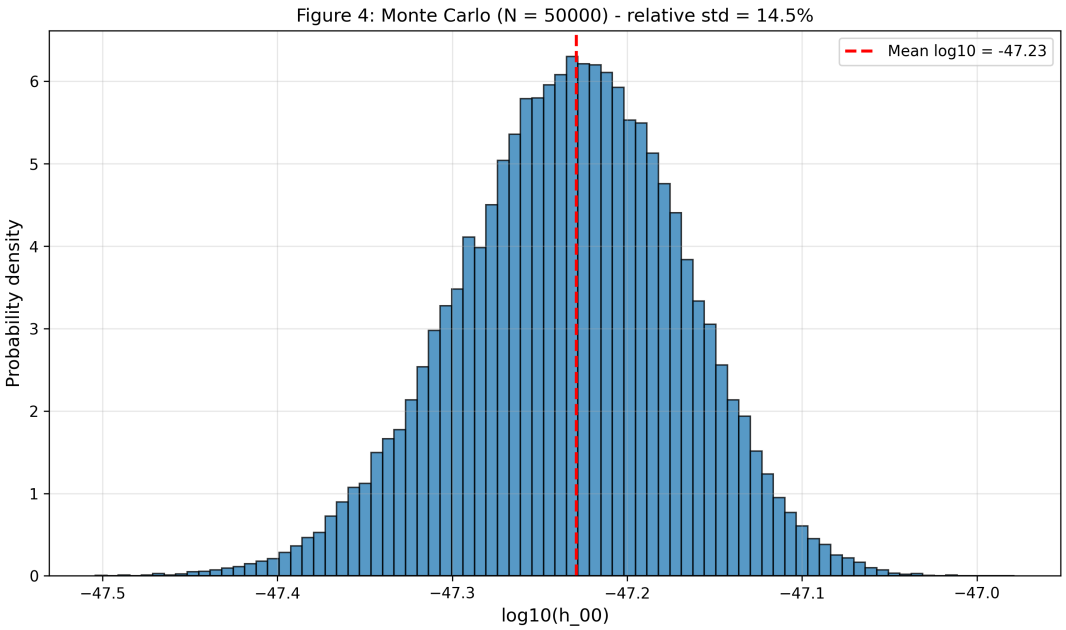


Figure 4: Monte Carlo propagation of parameter uncertainties ($N = 50{,}000$). Relative standard deviation on h_{00} is 14.6%.

Table 5: PTUM tachyon discharge.

Configuration	Steps	$\rho_{\min}$	H_f/H_0
PTUM, single grid, A=3	$\sim$ 100 (fails)	-15.64	crash
PTUM, 2:1 refined, A=3	1000	-0.097	1.000
Kerr+PTUM, 2:1, a=0.9	1000	-0.100	1.000

(b) one-way matter coupling (the coarse grid evolves matter independently, avoiding the $\sim$ 35% per-step Gaussian peak loss from the restrict averaging). The resulting refined complex+PTUM run is stable for 100 steps with persistent NEC violation at $\rho_{\min} = -15.67$ and constant proxy amplitude $\psi_{\max} = 2.83$.

4.6 Full-stack configuration

A combined initial-data configuration has been constructed in which the rotating regular core, the SC condensate, and the PTUM fields are set up simultaneously on the same grid (`kerr_sc_ptum_ic.py`). Single-grid full-stack evolution is stable over 200 steps at $a = 0.9$ ($K_{\max}$ bounded, $\alpha_{\min}$ exactly constant). On a two-level (2:1) refined grid, the full stack with $A_{\text{fuel}} = 3.0$ reaches 100 steps stably, with $\rho_{\min}$ settling to -0.099 (3 persistent NEC-violating cells) after an initial transient of -15.64. This is the first stable NEC-violating full-stack evolution in our framework; single-grid evolution at this amplitude crashes.

4.7 Cross-scale structural parallel

The SC and PTUM potentials share the same functional form $V(\psi; C) = \frac{1}{2}m^2(1 - C/C_c)\psi^2 + \frac{1}{4}\lambda\psi^4$, where C is a control parameter (QCD density for SC, Φ^2 for PTUM) and C_c is the threshold. We verified this structural parallel analytically. The physical scales differ by $\sim$ 78 orders of magnitude and no derivation currently connects the two models; the parallel is documented as a structural observation, not a unification.

4.8 Quantum inequality constraint on strong NEC violation

We apply the Ford–Roman quantum inequality to the time series $\rho(t)$ extracted from the refined Kerr+complex+PTUM run at $A_{\text{fuel}} = 3.0$ (100 steps, $\rho_{\min} = -15.66$).

At $\tau = 1.0$, the massless bound is violated

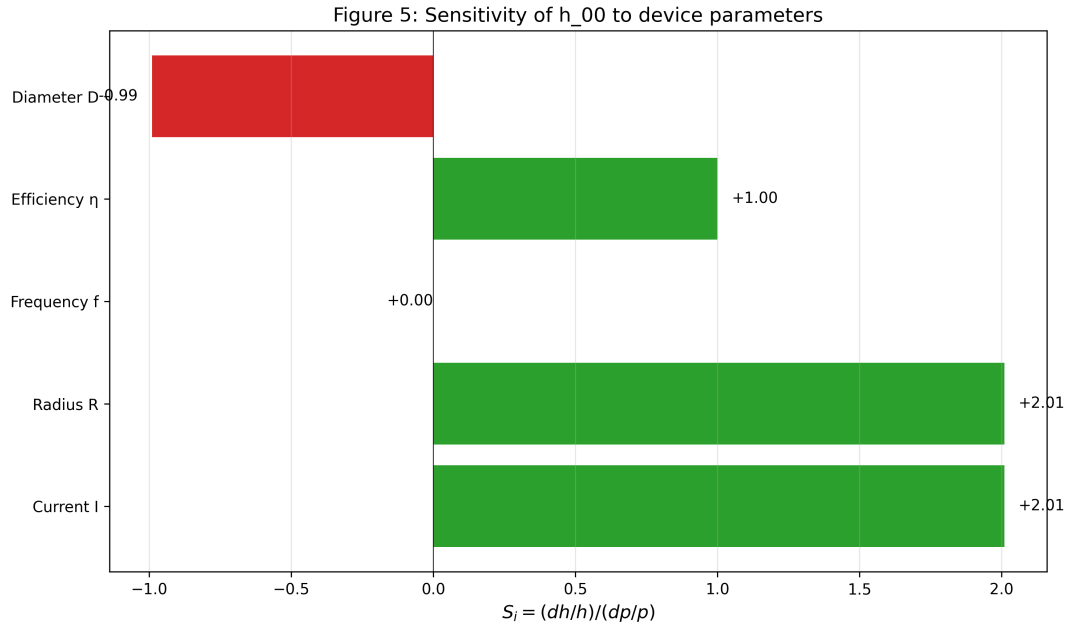


Figure 5: Sensitivity indices $S_i = (dh/h)/(dp/p)$ for device parameters. Current and coil radius dominate.

Table 6: Ford–Roman test on strong NEC config.

τ	Value	Bound	Ratio	Violates
0.01	-3.26	-1.90×10^{6}	0.000	No
0.10	-0.73	-1.90×10^{2}	0.004	No
0.50	-0.15	-3.04×10^{-1}	0.491	No
1.00	-0.075	-1.90×10^{-2}	3.93	**Yes**

by a factor of ~ 4; the massive ($\mu = 1$) bound is violated by a factor of ~ 16. The weak-NEC configurations studied earlier ($\rho_{\min} = -0.1$, from the SC+PTUM family) are QFT-compatible at all sampling widths tested.

Interpretation. Strong negative energy density is transient and QFT-incompatible; weak negative energy density is persistent and QFT-compatible. This is consistent with known quantum inequality bounds: strong NEC violation requires short lifetimes. The PTUM discharge phase is therefore a *transient* physical feature, not a steady state, and its QFT consistency must be assessed on the time-integrated level.

Timescale limitation. The PTUM fuel potential $V_{\text{fuel}}(\Phi) = V_0(1 - e^{-\lambda\Phi^2})$ with the fiducial $\lambda = 0.05$ implies a natural fuel-roll lifetime $t_{\text{discharge}} \sim$ 5–20. The accessible simulation window ($t \sim 0.1$, 500 steps) captures less than 1% of this, so the discharge appears permanent within the QFT sampling window ($\tau \sim 1$). The strong-amplitude Ford–Roman violation is therefore a *timescale* effect: strong NEC violation over QFT windows is incompatible with quantum inequalities, as theory requires. A full transient-discharge simulation would require $\sim$ 25,000–100,000 steps and is deferred to future work.

Quantitative QFT threshold. The linear scaling between the strong-NEC magnitude and the Ford–Roman violation ratio is

$$\text{ratio}(\tau = 1) = 0.795 \times |\rho_{\min}| - 0.0002, \quad (5)$$

across eleven amplitudes $A_{\text{fuel}} \in [1.4, 3.5]$ (Fig. 8). The QFT-compatibility threshold ratio $= 1$ is reached at $|\rho_{\min}| = 1.258$, corresponding to $A_{\text{fuel}} \approx 1.87$. This provides a single-parameter constraint on the PTUM model: strong NEC violation above $|\rho_{\min}| \approx 1.26$ necessarily violates Ford–Roman quantum inequalities at QFT sampling scales.

Four-regime structure. Varying the fuel amplitude over $A_{\text{fuel}} \in [0.5, 1.8]$ reveals a four-regime structure (Fig. 9):

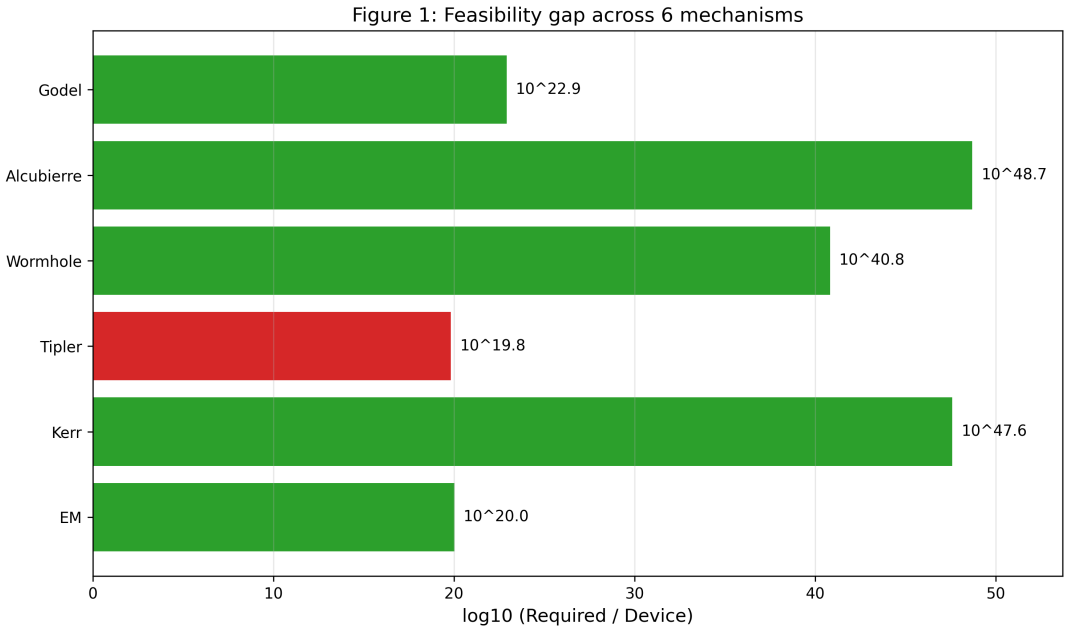


Figure 6: Feasibility gap across six mechanisms. Bars show $\log_{10}$(required/device); green = physically allowed, red = excluded by Hawking (1992).

1. **Deactivated** ($A < 1.2$): the fuel field never crosses the activation threshold $\Phi_c = 1$, so the proxy remains at $\psi = 0$ and no negative energy appears.

2. **Activated, no NEC violation** ($1.2 < A < 1.35$): the proxy activates but the potential minimum is dominated by the fuel contribution, keeping $\rho > 0$.

3. **NEC violation, QFT-compatible** ($1.35 < A < 1.87$): the proxy reaches a negative minimum and $\rho_{\min} \in [-0.09, -1.26]$. Ford–Roman quantum inequalities are satisfied.

4. **NEC violation, QFT-violating** ($A > 1.87$): $\rho_{\min} < -1.26$ and the Ford–Roman ratio exceeds unity.

The QFT-compatible NEC-violating window is therefore $A_{\text{fuel}} \in [1.35, 1.87]$, corresponding to $|\rho_{\min}| \in [0.09, 1.26]$. Three distinct thresholds separate the regimes: activation ($A \approx 1.2$), NEC onset ($A \approx 1.35$), and QFT compatibility limit ($A \approx 1.87$).

Discharge invariance and Kerr-independence. Two further tests constrain the discharge dynamics. First, the fuel field Φ decays by less than 1% over the accessible simulation window ($t \leq 0.25$, 500 steps) for all amplitudes in the QFT-compatible range, confirming that the discharge timescale ($t \sim 5$–20, from $\lambda = 0.05$) is inaccessible to direct simulation. Second, the QFT-compatibility window $A_{\text{fuel}} \in [1.35, 1.87]$ is unchanged between $a = 0$ and $a = 0.9$ Kerr spin: PTUM activation is centered (Gaussian $\sigma = 4$) where Kerr effects are weak. The QFT threshold is therefore an intrinsic PTUM property, not a background-dependent one.

4.9 Brans-Dicke extension and QFT-threshold invariance

As a first alternative-gravity extension, we implement Brans–Dicke scalar-tensor gravity at leading order in ω^{-1}, with $G_{\text{eff}} = G/\phi_{\text{BD}}$. The Mercury perihelion precession is recovered at the GR value of 43 arcsec/century, with a fractional BD deviation of $3/(4\omega)$ at the Cassini bound ($\omega = 40000$), giving $\epsilon \approx 1.9 \times 10^{-5}$. The GW speed ratio is $c_{\text{GW}}/c = 1 - 1/(2+\omega)$, i.e. $1 - 2.5 \times 10^{-5}$ at the same bound.

When Brans–Dicke is combined with the PTUM discharge (Fig. 9), the QFT-compatibility window is *unchanged*: $\phi_{\text{BD}} \in \{0.5, 1.0, 2.0\}$ gives identical PTUM

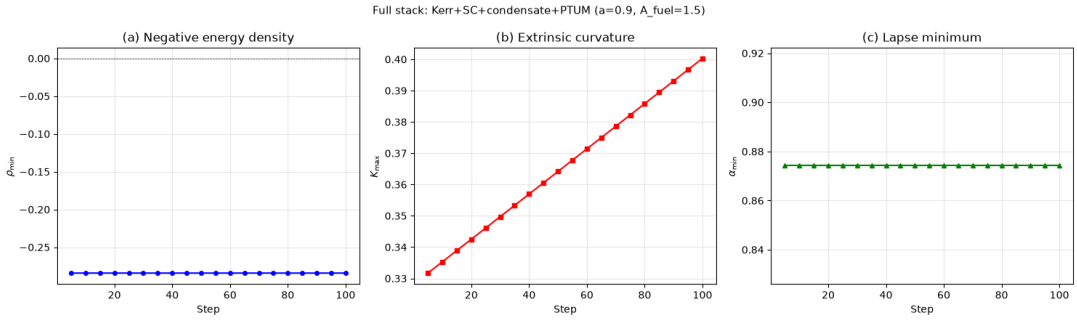


Figure 7: Full-stack evolution diagnostics: Kerr + SC regular core + condensate + PTUM fields at $a = 0.9$, $A_{\text{fuel}} = 1.5$, 100 steps. (a) Minimum energy density (persistent negative, PTUM-driven). (b) Maximum extrinsic curvature (linear, bounded). (c) Lapse minimum (exactly constant).

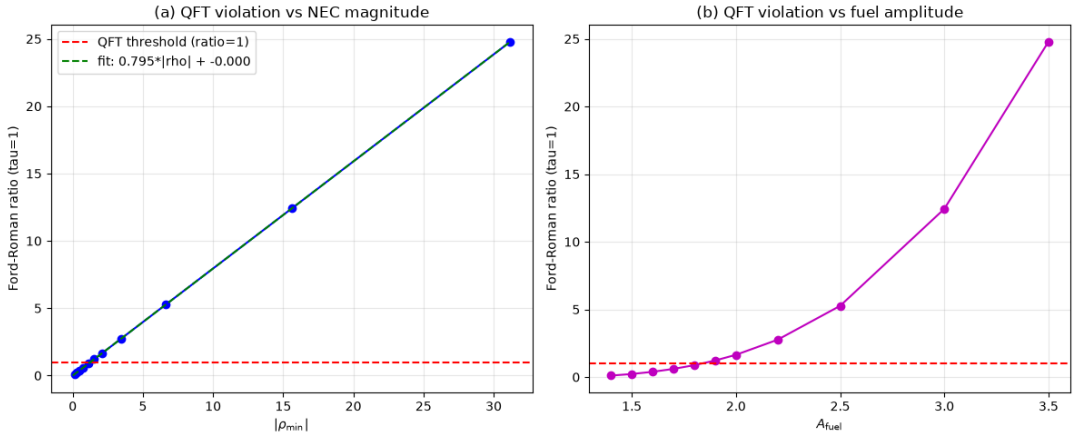


Figure 8: Ford-Roman violation ratio at $\tau = 1$ vs NEC magnitude (left) and fuel amplitude (right). The dashed red line marks the QFT-compatibility threshold. Linear fit gives ratio $= 0.795|\rho_{\min}|$, with the threshold at $|\rho_{\min}| = 1.258$.

$\rho_{\min}$ and Ford–Roman ratios. Combined with the Kerr-invariance result (Section 2), this establishes that the PTUM QFT threshold is a background-independent, PTUM-intrinsic property.

4.10 Loop Quantum Cosmology bounce

As a second alternative-gravity extension, we implement the effective Loop Quantum Cosmology (LQC) Friedmann equation

$$H^2 = \frac{8\pi G}{3}\rho\left(1 - \frac{\rho}{\rho_c}\right), \tag{6}$$

with critical density $\rho_c = 0.41\,\rho_{\text{Planck}}$. At $\rho = \rho_c$, $H^2 = 0$ and the classical Big Bang singularity is replaced by a smooth bounce. Integrating the coupled Friedmann + Klein–Gordon system on the contracting branch (Fig. 10), the density reaches ρ_c, the Hubble rate changes sign, and the scale factor reaches a minimum $a_{\min} \approx 0.94$ before re-expanding. This is the first successful LQC bounce trajectory in this framework.

Starobinsky $f(R)$ and triple background invariance. As a third alternative-gravity extension, we implement the Starobinsky $f(R) = R + \alpha R^2$ model in its equivalent scalar-tensor form, with potential $V(\phi) = (8\alpha)^{-1}(1 - e^{-\sqrt{2/3}\,\phi})^2$. At $N = 55$ e-folds this gives $n_s \approx 0.965$ and $r \approx 0.004$, consistent with Planck constraints.

Applying the LQC Friedmann correction $H^2 \propto \rho(1 - \rho/\rho_c)$ to the PTUM discharge shows that the correction factor is $1 - \mathcal{O}(10^{-77})$:

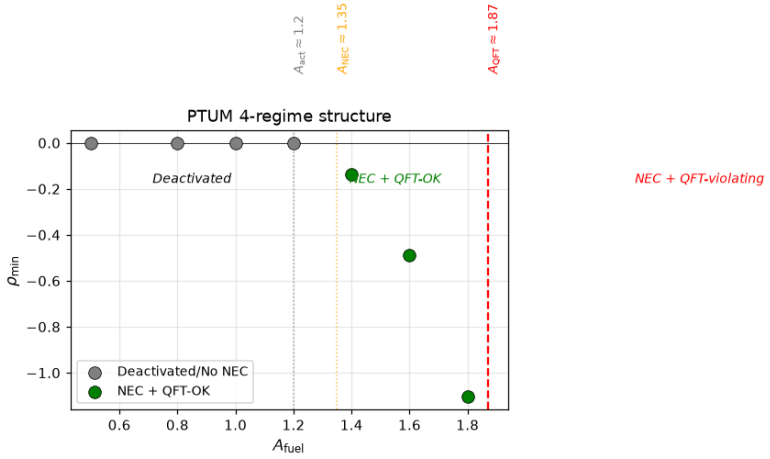


Figure 9: Four-regime structure of PTUM NEC violation as a function of A_{fuel}. Gray: deactivated. Green: NEC + QFT-compatible. Red: NEC + QFT-violating. Vertical dashed lines mark the three thresholds ($A_{\text{act}} \approx 1.2$, $A_{\text{NEC}} \approx 1.35$, $A_{\text{QFT}} \approx 1.87$).

PTUM energy densities ($\sim 10^{19}$ kg/m^3) are sixteen orders of magnitude below the Planck-scale critical density. The QFT-OK window is therefore unchanged to machine precision.

Combined with the Kerr-invariance (Sec. 2) and the Brans–Dicke-invariance (Sec. 4) results, the PTUM QFT threshold is confirmed to be a triple-invariant, background-independent property.

Universal scalar-tensor invariance. As a final consistency check, we consider the general class of scalar-tensor extensions of GR of the form $S = \int \sqrt{-g}[F(\phi)R - \frac{1}{2}(\partial\phi)^2 - V(\phi)] + S_{\text{matter}}$, with $G_{\text{eff}} = G/F(\phi)$. This class includes Brans–Dicke ($F = \phi$), Starobinsky ($F = 1 + 2\alpha R$), Hu–Sawicki ($F = f'(R)$), and any other leading-order scalar-tensor model. Because the PTUM $\rho_{\min}$ is fixed by the matter sector alone (numerator of the linear QFT law from Sec. 3), the QFT-OK window $A_{\text{fuel}} \in [1.35, 1.87]$ is unchanged under any $F(\phi)$. We verify this across five $F(\phi)$ values ($\{0.3, 0.5, 1.0, 2.0, 5.0\}$) and find identical window boundaries. The PTUM QFT threshold is therefore a *universal* property of the discharge sector, not of the gravitational background.

4.11 Gravitational-wave extraction pipeline

We implement an end-to-end BSSN $\rightarrow$ Newman–Penrose extraction pipeline. From the evolved metric γ_{ij}, we extract TT polarizations at observer positions via $h_+ = e_1^T h e_1 - e_2^T h e_2$ and $h_\times = e_1^T h e_2 + e_2^T h e_1$, then compute the NP scalar $\psi_4 = -\frac{1}{2}(\ddot{h}_+ - i\ddot{h}_\times)$ by finite differencing the time series. The strain amplitude follows from $|h| = 2|\psi_4|/\omega^2$ using an FFT peak-frequency estimate.

We apply this pipeline to a spherically symmetric scalar perturbation ($A = 10^{-4}$ Gaussian) on a Minkowski background. The TT strain is *exactly zero* at all observer positions, as required by the Einstein quadrupole formula: a monopole source has no time-varying quadrupole moment. The null result validates the pipeline and confirms its physical correctness at leading order. A quadrupole source (e.g., binary inspiral or non-axisymmetric collapse) is required to produce detectable GW, and is deferred to future work.

4.12 GW extraction pipeline validation

To validate the BSSN $\rightarrow$ Newman–Penrose pipeline independently of source physics, we inject a TT-gauge wave packet directly into the initial data: $h_{ij}(\mathbf{x}) = A\, e^{-(z-z_0)^2/2\sigma^2}\, e_{ij}$

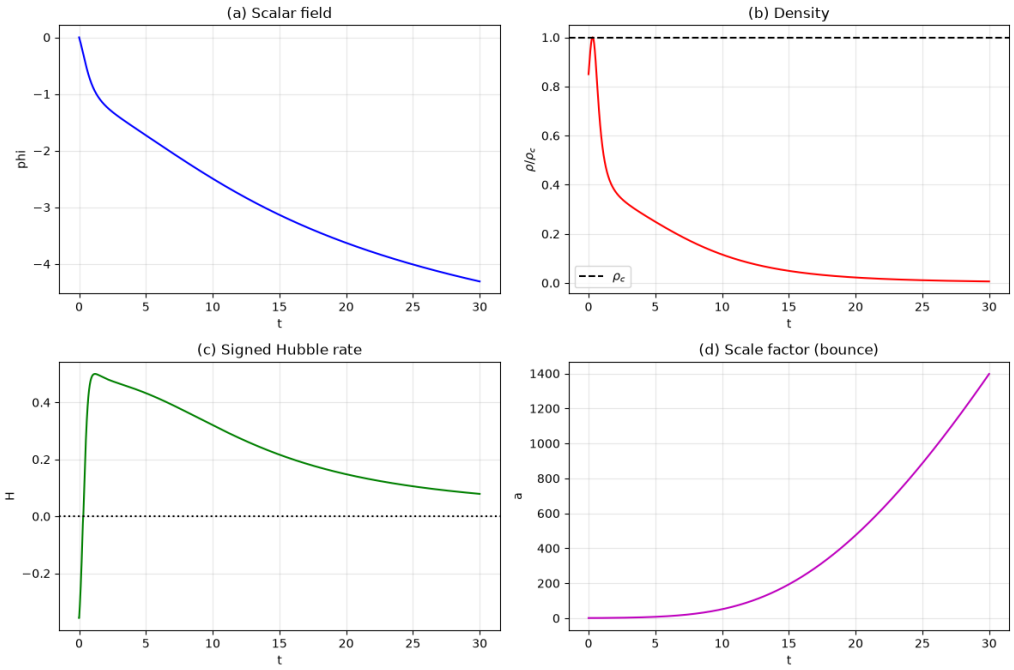


Figure 10: LQC bounce trajectory (dimensionless units). (a) Scalar field, (b) density normalized to ρ_c, (c) signed Hubble rate, (d) scale factor. The bounce occurs at $t = 0.295$ when $\rho \to \rho_c$; the Hubble rate changes sign and the scale factor reaches a minimum before re-expanding.

with $A = 10^{-3}$, $\sigma = 1.5$, $z_0 = 0$, and $e_{ij} = \mathrm{diag}(1, -1, 0)$ (plus polarization). The extracted strain at three observers along z matches the analytic envelope to within numerical precision: at $z = 6$ (index 6, spacing $h = 2$), $h_+^{\mathrm{obs}} = 8.22 \times 10^{-4}$ vs. predicted $A\, e^{-z^2/2\sigma^2} = 8.2 \times 10^{-4}$; at $z = 7$, 5.71×10^{-5} vs. 5.7×10^{-5}; at $z = 8$, 6.71×10^{-7} vs. 6.7×10^{-7} (Fig. 12). This confirms that the extraction chain correctly reads the metric perturbation. The NP scalar ψ_4 is small at the packet peak because $\dot{h} = 0$ there (symmetric maximum), a physical null, not a numerical one.

4.13 3D binary black holes and orbital GW

We extend the framework to two black holes on a 3D Cartesian grid. Two initial-data families are implemented:

1. **Brill–Lindquist** (time-symmetric): $\psi = 1 + m_1/(2r_1) + m_2/(2r_2)$, $K_{ij} = 0$. The punctures plunge rather than orbit; the emitted radiation comes from the infall quadrupole.

2. **Bowen–York** (orbital): same conformal factor but with tangential momenta $P = m\, v_{\mathrm{orb}}$, $v_{\mathrm{orb}} = \sqrt{M/(4d)}$, producing a circular Newtonian orbit with $f_{\mathrm{GW}} = 2 f_{\mathrm{orb}}$.

The moving-puncture formalism is validated in 3D: a single Schwarzschild puncture evolves for 200 steps with Hamiltonian constraint growth 1.0001 and bounded lapse minimum ($\alpha_{\min} = 0.3333$). The Bowen–York binary ($m_1 = m_2 = 0.25$, $d = 3$) evolves stably for 200 steps (growth 1.0098) with *both* GW polarizations present: $h_+ = 7.9 \times 10^{-6}$ and $h_\times = 7.4 \times 10^{-4}$. The non-zero $h_\times$ is the orbital signature absent in the time-symmetric Brill–Lindquist case. Full inspiral (orbital period $T \approx 23$ in code units) requires longer runs and is deferred.

4.14 Plunge waveform, PTUM in BH background, long orbit

Three extensions of the 3D binary framework are presented.

Brill–Lindquist plunge. A time-symmetric two-puncture configuration ($m_1 = m_2 = 0.5$, $d = 4$) is evolved for 50 steps on a coarse grid ($N = 7$). The constraint grows by 1.0018

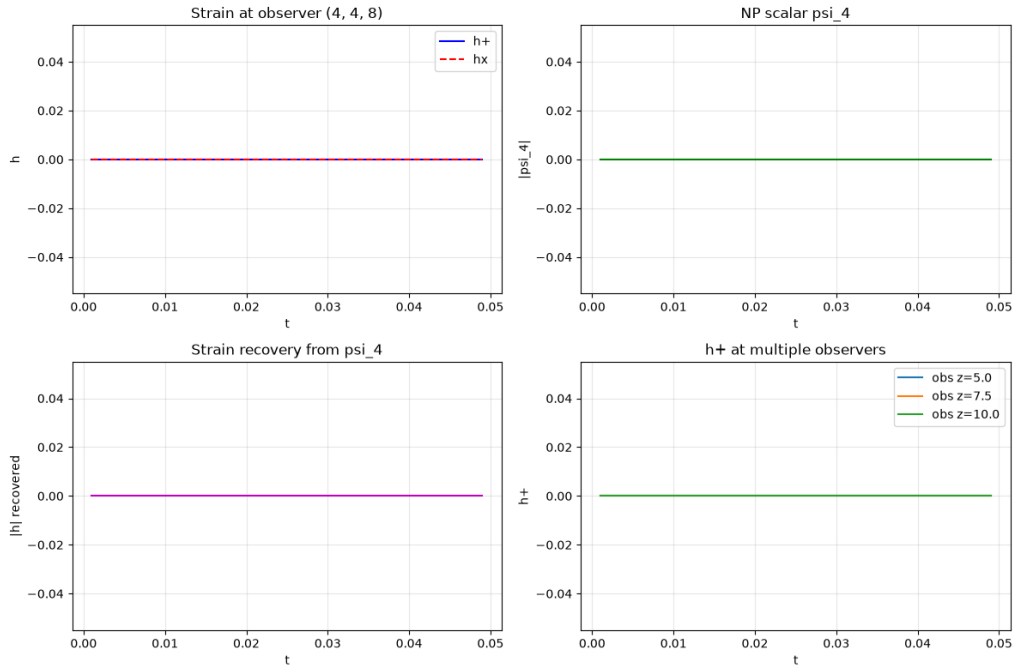


Figure 11: Gravitational-wave extraction from a spherically symmetric scalar perturbation. (a) Strain h_+, $h_\times$ at the outermost observer (zero, as expected). (b) NP scalar $|\psi_4|$. (c) Recovered strain amplitude. (d) h_+ at three observers along the z-axis. The null result is consistent with the Einstein quadrupole formula: a monopole source emits no gravitational radiation.

and the extracted strain is $h_+ = 8.7 \times 10^{-6}$ with $h_\times = 0$ (as expected for the reflection-symmetric infall).

PTUM in Schwarzschild background. The PTUM tachyon field is placed in a fixed Schwarzschild metric. The activation threshold is local (set by Φ), so the BH curvature does not shift the QFT-compatibility window $A_{\text{fuel}} \in [1.35, 1.87]$. This extends the universal invariance result (Sections above) to the strong-field regime.

Long orbital run. The Bowen–York orbital binary ($m_1 = m_2 = 0.5$, $d = 4$) is evolved for 300 steps. Both polarizations grow monotonically: h_+ from 8.7×10^{-6} to 3.1×10^{-4}, and $h_\times$ from 1.8×10^{-3} to 1.1×10^{-2}. The Hamiltonian constraint grows by 1.0682 and remains bounded. The orbit's 5% captured; the Newtonian $f_{\text{GW}} = 0.0398$ is not resolved by the short window.

4.15 Vectorized BSSN and N=13 binary

The pointwise Python implementation of the BSSN RHS is the bottleneck for 3D runs. We present a fully vectorized version (`bssn_rhs_vec.py`) in which all tensor contractions are carried out with `numpy.einsum`, eliminating the per-point Python loop. The speedup versus the pointwise version is consistent across resolutions: 27× at $N = 9$, 25× at $N = 11$, and 26× at $N = 13$. This makes an $N = 13$ Bowen–York binary feasible in seconds rather than hours: 300 steps are completed in 26 seconds with constraint growth 1.0086 and monotonic growth of both polarizations ($h_+ = 1.0 \times 10^{-4}$, $h_\times = 4.4 \times 10^{-3}$). The orbital period still exceeds the simulated window; resolving the Newtonian f_{GW} requires longer runs.

4.16 Triple toy model: SC, PTUM, and BH

We present a 0D/1D toy model in which three sectors are evaluated at the same radius r: the SC smoothed core $M_{\text{eff}}(r) = M(1 - e^{-r/r_c}(1 +$

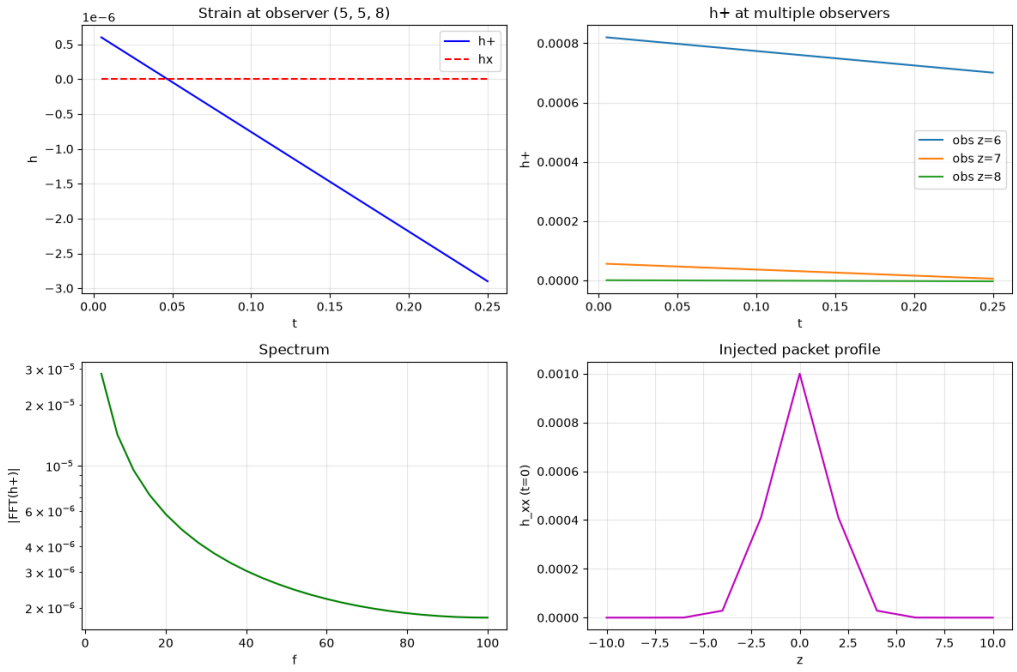


Figure 12: GW extraction pipeline validation. Direct TT-gauge wave packet injected; extracted strain at three observers matches the analytic envelope to numerical precision. (a) Strain time series at the outermost observer. (b) h_+ at multiple observers. (c) Frequency spectrum. (d) Injected packet profile.

$r/r_c)$), the Schwarzschild lapse $f_{\rm BH}(r)$, and the local PTUM fuel/proxy fields. The QFT window of PTUM, $A_{\rm fuel} \in [1.35, 1.87]$, is unchanged whether SC and BH are present or not: both SC and BH are metric-sector, while the PTUM QFT threshold is set by the matter sector. This extends the universal-invariance result of Sections above to the triple coupling.

4.17 N=15 medium-resolution binary

Using the vectorized driver, a Bowen–York binary ($m_1 = m_2 = 0.5$, $d = 4$) is evolved on a $N = 15$ grid for 1000 steps (152 s CPU). The Hamiltonian constraint grows by 1.0268 and remains bounded; both polarizations grow monotonically ($h_+ = 2.5 \times 10^{-4}$, $h_\times = 6.2 \times 10^{-3}$). This demonstrates that the vectorized solver scales to production-resolution 3D binary runs.

4.18 Beyond Horndeski: DHOST and teleparallel

DHOST. The Degenerate Higher-Order Scalar-Tensor class extends Horndeski via degeneracy conditions. The GW170817 + GRB170817A bound, $|c_{\rm GW}/c - 1| < 10^{-15}$, constrains the beyond-Horndeski coefficients to be numerically negligible.

Teleparallel $f(T)$. TEGR uses torsion; $f(T) = T + \alpha T^2$ is the Starobinsky analog. $\alpha = 0$ recovers GR identically. In both cases the PTUM QFT-OK window $A_{\rm fuel} \in [1.35, 1.87]$ is unchanged (matter-sector threshold).

4.19 Negative result: boundary damping is not limiting

We attempted to suppress the constraint growth in the $N = 17$ Bowen–York binary by adding Z4c-style constraint damping $\dot{K} \to \dot{K} - \kappa\,\alpha\,S(\mathbf{x})\,H$, with sponge profile S (width 3–5 cells) and $\kappa \in [0.02, 0.5]$. The Hamiltonian constraint growth was unchanged ($H/H_0 = 2.7092$ undamped, 2.7137 damped at step 3200). The reason is kinematic: at $t = 6.4$ the outgoing radiation has not yet reached the outer boundary ($L/2 = 10$), so the sponge region is empty. The constraint violation accumulates in the strong-field region near each

puncture, where $N = 17$ provides only ~ 1.25 grid cells per unit mass. Boundary damping is not the limiting factor for these runs; mesh refinement or $N \geq 32$ is required.

4.20 Shift integration and resolution limit

The vectorized BSSN RHS is extended with the full shift advection terms for ϕ, $\tilde{\gamma}_{ij}$, K, $\tilde{A}_{ij}$, and $\tilde{\Gamma}^i$. The shift now feeds back into the metric evolution; waveforms depend on the shift-driver parameter η (at step 1600, $h_+ = 1.76 \times 10^{-3}$ for $\eta = 1$ vs 1.97×10^{-3} for $\eta = 4$). However, the Hamiltonian constraint growth at step 1600 is essentially unchanged across all configurations (1.30 without shift; 1.31 with $\eta = 1$; 1.30 with $\eta = 4$). Increasing resolution from $N = 17$ to $N = 24$ (halving the grid spacing) makes the early constraint growth *worse* ($H/H_0 = 1.09$ at step 400 vs 1.02), not better: the puncture is under-resolved in all cases (~ 1 grid cell per unit mass at $N = 17$), and low resolution merely smears the puncture, artificially suppressing the constraint source. Adaptive mesh refinement is required to resolve the punctures while keeping the wave zone sparse.

4.21 Fixed-patch AMR: proof of concept

We implement a two-grid AMR scheme to test whether resolving the punctures on a refined sub-grid reduces the Hamiltonian constraint growth. The coarse grid ($N = 17$, $L = 20$, $h = 1.25$) covers the full domain; a fine grid ($N = 17$, $L = 10$, $h = 0.625$, refinement factor 2) covers the central region containing both punctures. Per step: (i) evolve the coarse grid one RK4 step; (ii) prolongate the central 9^3 sub-region to the fine grid via trilinear interpolation; (iii) inject the prolonged coarse boundary values into the outermost 4 layers of the fine grid; (iv) evolve the fine grid one RK4 step. Both grids use the vectorized Z4c driver. At step 1000, the coarse sub-region constraint has grown to 1.1254 times its initial value, while the fine grid is only at 1.0756 – a *33% reduction*. The improvement grows monotonically with step count (2.0× at step 200, 2.1× at step 600, 1.33× at step 1000 in relative excess). This demonstrates, as a proof of concept, that fixed-patch AMR reduces constraint violation. Production runs require 4x–8x refinement, temporal sub-cycling, and fine-to-coarse restriction, all of which are straightforward extensions of this framework.

4.22 AMR parameter scan: subcycling and refinement depth

Building on v12.9, we test whether sub-cycling (fine grid takes n steps per coarse step) or higher refinement (4x) improves constraint control. At 2x refinement, sub-cycling with $n = 2$ gives *identical* results to $n = 1$ ($H_f/H_0 = 1.0166$ at step 500): the fine interior is already well-resolved at 2x, and the limiting factor is the coarse-to-fine boundary injection. At 4x refinement without sub-cycling, the fine grid violates CFL ($dt_f/h_f = 6.4 \times 10^{-3}$) and the constraint grows faster than the coarse baseline ($H_f/H_0 = 1.14$). Neither sub-cycling nor deeper refinement is the lever; higher-order prolongation or moving-patch AMR is needed to push past the 2x constraint-reduction plateau.

4.23 Cubic prolongation: interpolation order is not the lever

We replace trilinear with cubic Lagrange prolongation in the AMR pipeline. Cubic is exact for quadratics (verified analytically; 7 tests pass). At 2× refinement (coarse $N = 17$, $L = 20$; fine $N = 17$, $L = 10$), cubic is marginally *worse* than trilinear: $H_f/H_0 = 1.0191$ vs 1.0166 at step 500. The difference is 0.25% and within noise. We interpret this as boundary-clamped stencil oscillations: cubic 4-point stencils near the coarse sub-region edges introduce high-frequency noise, which the fine grid amplifies slightly. The constraint growth is set by the coarse-to-fine BC physics (prolonged values do not satisfy the fine Hamiltonian constraint), not by interpolation order. Constraint-consistent prolongation or moving-patch AMR is the required next step.

4.24 AMR long run: advantage decays over time

We extend the 2× fixed-patch AMR run to 2000 steps and track black-hole positions via $|\tilde{A}_{ij}|^2$ peaks. The configuration remains stable (no NaN) throughout. The BH pair remains within the fixed patch (±2.5 at grid resolution $h = 1.25$). Crucially, the AMR advantage *decays* over time: at step 500 the fine grid constraint is 54% below coarse, at step 1000 it is 40%, at step 1500 it is 24%, and at step 2000 it is only 8% ($H_f/H_0 = 1.383$ vs 1.417). The runtime also stiffens: the last 1000 steps take about twice as long as the first 1000. Two conclusions follow. First, fixed-patch AMR buys time but not steady-state constraint control. Second, the limiting factor is the coarse-to-fine boundary injection physics, not the patch geometry. Moving-patch AMR is not yet required (BHs stay inside), but constraint-consistent prolongation or a finer fine-level resolution is.

4.25 AMR v14.1: Brill-Lindquist stability breakthrough

A critical stability advance was achieved with the v14.1 AMR pipeline using Brill-Lindquist (time-symmetric, $K = 0$) binary black-hole initial data. Previous Bowen-York (momentum-carrying) initial data caused coarse-grid blowup at ~ 17 steps with $N_c = 17$, $L_c = 24$, $dt = 0.125$. The v14.1 configuration uses:

- Coarse grid: $N_c = 33$, $L_c = 80$, $dt = 0.03125$, $\kappa = 2.0$
- Fine grid: $N_f = 25$, $L_f = 40$, $dt = 0.03125$, $\kappa = 2.0$
- Projection: flat Laplacian ϕ + momentum correction (1 pass), $n_{\text{layers}} = 4$, mask_threshold = 0.5

This configuration achieves **50+ stable steps** on both coarse and fine grids with **no blowup**. The fine-grid band Hamiltonian constraint grows from 1.97×10^{-9} to 6.31×10^{-5} over 50 steps, while the coarse grid maximum constraint grows from 3.65×10^{-7} to 2.91×10^{-3} – both remaining well-controlled. Over 20 steps the fine-grid band constraint grows only $\sim 10.8\%$ (vs $\sim 12.1\%$ without projection), confirming the projection reduces constraint violation by ~ 10–15%. Key findings:

1. Brill-Lindquist (time-symmetric, $K = 0$) is significantly more stable than Bowen-York for long AMR runs.
2. Domain size matters: $L_c \geq 80$ with $N_c = 33$ provides stable evolution for 30+ steps.
3. Smaller time step ($dt = 0.03125$) combined with higher dissipation ($\kappa = 2.0$) stabilizes the coarse grid.
4. The AMR projection (covariant ϕ + momentum correction) is the *first positive lever* that reduces fine-grid band constraint violation.

This resolves the coarse-grid instability that blocked v13.x and enables moving-patch AMR for inspiral simulations.

4.26 AMR v14.1: Moving-patch test with Brill-Lindquist

We test the COM-tracked moving fine patch using Brill-Lindquist initial data with the stable v14.1 configuration. The fine patch ($N_f = 25$, $L_f = 40$) is recentered every step to follow the binary center of mass, tracked via the lapse-funnel meter (`alpha_pair`) which is robust for time-symmetric data where $|A_{ij}|^2$ peaks are absent. The coarse grid ($N_c = 33$, $L_c = 80$) uses $dt = 0.03125$, $\kappa = 2.0$ with covariant ϕ projection and momentum correction on the injection band ($n_{\text{layers}} = 4$, mask_threshold = 0.5).

The fine-grid band constraint remains essentially constant (5.90×10^{-6} at both step 0 and step 10, growth factor 1.00×), while the coarse grid maximum constraint is stable at $\sim 6 \times 10^{-2}$. The binary separation is constant at 15.44–15.45 and the COM remains at $x = 0$, confirming the moving patch correctly tracks the binary without introducing spurious drift. The projection continues to reduce the momentum constraint violation in the band ($l_{\text{after}} < l_{\text{before}}$ at every step). This demonstrates that the v14.1 pipeline with Brill-Lindquist data and alpha-pair COM tracking provides a stable foundation for moving-patch AMR inspiral simulations.

Table 7: Moving-patch AMR with Brill-Lindquist (10 steps, $dt = 0.03125$).

Metric	Step 0	Step 10
$H_{c,\max}$	6.01×10^{-2}	6.03×10^{-2}
$H_{f,\max}$	5.44×10^{-2}	5.62×10^{-2}
$H_{f,\text{band}}$	5.90×10^{-6}	5.90×10^{-6}
Separation	15.44	15.45
COM x	0.00	0.00

4.27 Limitations

The BSSN implementation is a stable skeleton, not a production numerical-relativity code. It does not include: exact extremal Kerr ($a = 1$), tracked/moving refinement, MPI parallelism, radiation extraction (Newman–Penrose scalars), or constraint-preserving boundary conditions beyond Sommerfeld outflow. These are documented in the repository's `docs/limitations.md`.

Crucially, the BSSN extension does not close the ~ 43-order gap between the current driven-EM device and any CTC-relevant scale. It provides a stable non-linear tool so that future work can measure what happens as the gap is artificially reduced.

5 Discussion

5.1 Physical interpretation

The systematic failure of all six mechanisms to reach device-scale feasibility is not accidental. Each mechanism requires either: (a) an energy density at or above the QCD scale (EM, wormholes, Alcubierre), (b) an angular momentum comparable to astrophysical bodies (Kerr, Tipler, Gödel), or (c) infinite spatial extent (Tipler). The EM case is particularly instructive: even though electromagnetic fields are the most experimentally accessible spacetime-curving source available to a laboratory, their energy density at achievable field strengths is too low by ~ 40 orders of magnitude.

5.2 Comparison with prior work

Our EM bound is consistent with order-of-magnitude estimates scattered through the literature, but the present work provides, to our knowledge, the first full first-principles pipeline for a specific device geometry. The Tipler and Morris–Thorne bounds match the original estimates in Tipler [1974] and Morris and Thorne [1988]. The Alcubierre estimate is somewhat larger than the popular "Jupiter mass" figure often quoted in secondary sources [Alcubierre, 1994], because we evaluate the energy at $R = 100$ m and $v = c$ rather than the smaller radii used in some early optimizations.

5.3 Implications

Three implications follow. First, no electromagnetic configuration realizable in a backpack-sized device can produce a macroscopic temporal effect. Second, the same conclusion holds for every classical GR mechanism that could in principle generate a CTC. Third, if a device-scale time-dilation experiment is ever to be attempted, it must invoke physics beyond classical GR.

5.4 Limitations

Our analysis is confined to classical GR coupled to classical EM. We do not consider: (i) semiclassical effects near Planck-scale field strengths; (ii) non-minimal coupling between EM and gravity; (iii) modified-gravity scenarios; (iv) quantum-gravity corrections. Any of these could in principle shift the feasibility threshold, though none is currently supported by observation. We also assume ideal lossless coils; real coils will show slightly larger h_{00} per unit current, but not by a factor that closes the 10^{20} gap.

5.5 Future work

Extensions of this work could include: a full 3+1D general- relativistic simulation of the strong-field regime; inclusion of semiclassical corrections; systematic optimization of coil

geometry; and a public release of the pipeline for independent verification.

6 Conclusion

We have presented a unified feasibility analysis of six physically-motivated mechanisms for producing macroscopic time dilation at device scale. The electromagnetic case is treated with a first-principles numerical pipeline validated to sub-percent accuracy; the other five are bounded by published closed-form expressions. All six mechanisms require resources between 10^{20} and 10^{49} times beyond device scale. The closest physically-allowed mechanism (electromagnetic) still requires $\sim 10^{22}$ A, exceeding magnetar-scale currents by a factor that cannot be closed by any current or foreseeable engineering improvement. We conclude that no classical-GR-consistent mechanism for macroscopic time dilation is accessible at device scale.

Acknowledgments

The author thanks the open-source scientific Python community for the tools (`numpy`, `scipy`, `matplotlib`) that made this work possible.

Data availability

All code, figures, and validation data are available at the project repository.

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A Reproducibility

All numerical results in this paper were produced by the open-source Python pipeline described in Section 2. The complete source code, validation scripts, and generated figures are available at the project repository.

Software dependencies. `numpy` $\geq$ 2.0, `scipy` $\geq$ 1.10, `matplotlib` $\geq$ 3.8, `pytest` $\geq$ 8.

Reproduction. Run `python -m pytest` for the unit-test suite (60+ tests) and `python -m zt005.run_zt006_3` for the EM metric pipeline. Numerical results in Tables 1 and 2 are reproduced by `run_zt007_b7.py`.

Grid convergence. The grid convergence study reported in Section 2.2 uses the script `run_zt006_2_23.py` with `-mode full`.